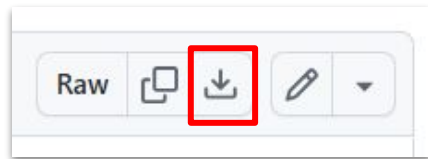


Lesson 4: Gridded Data with xarray

As you arrive and get set up...



1. Open **CryoCloud**: hub.cryointhecloud.com
 - Start a 1.9GB Python node
 - Create a folder named `lessons` (or any way you want to organize it)



2. Make a copy of **today's lesson notebook**:
 - In your web browser, visit:
https://github.com/NASA-SARP/2026-Coding-Alex/blob/main/lessons/04_gridded_data.zip
 - Download the zip folder `04_gridded_data.zip` and unzip:
 - “Drag and drop” the unzipped folder `04_gridded_data` into the `lessons` folder you created earlier in CryoCloud

Recap: From data types to data structures

Data types

Data type	Python name	What it holds	Examples
Integer	int	Whole numbers	14, 0, -3
Float	float	Decimal numbers	3.5, -0.2, 6.022e23
String	str	Text	"Jack", 'hello'
Boolean	bool	True or False	True, False

Data structures

List

```
mixed = ['abc', 34, True, 40, 'male']
```

DataFrame

	temperature	humidity	wind_speed
0	25	60	10
1	20	65	9
2	27	62	10
3	31	80	6

Dictionary

```
metadata = {  
    'location': 'Nairobi',  
    'quality_flag': 5,  
    'zenith': 60,  
    'clouds': True,  
}
```

GeoDataFrame

	COUNTYFP	NAME	geometry
0	001	Alameda	POLYGON ((-13612246.763 4538150.085, -13612346...
1	003	Alpine	POLYGON ((-13366503.508 4678946.015, -13366492...
2	005	Amador	POLYGON ((-13472697.863 4647652.232, -13472697...
3	007	Butte	POLYGON ((-13565004.793 4798394.124, -13564991...
4	009	Calaveras	POLYGON ((-13428575.483 4627725.227, -13428534...

Xarray DataArray

```
xarray.DataArray (latitude: 5, longitude: 6)
```

```
array([[80, 80, 81, 83, 82, 80],  
       [81, 80, 81, 83, 83, 82],  
       [81, 81, 83, 84, 84, 83],  
       [79, 80, 83, 85, 85, 85],  
       [78, 77, 80, 85, 86, 87]])
```

▼ Coordinates:

latitude (latitude) int64 36 37 38 39 40

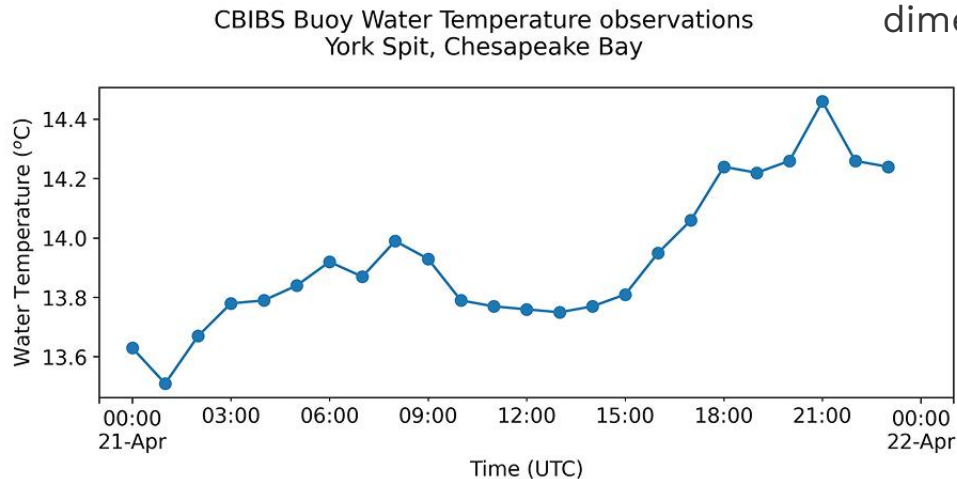
longitude (longitude) int64 -80 -79 -78 -77 -76 -75

► Indexes: (2)

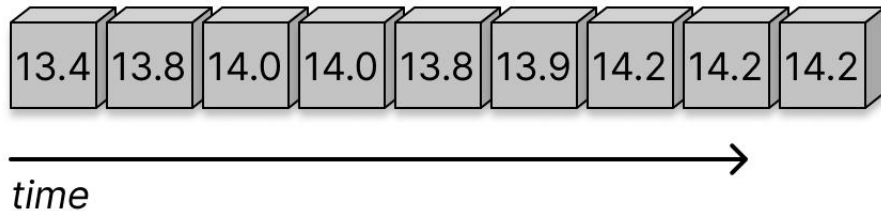
► Attributes: (0)

Multi-dimensional data

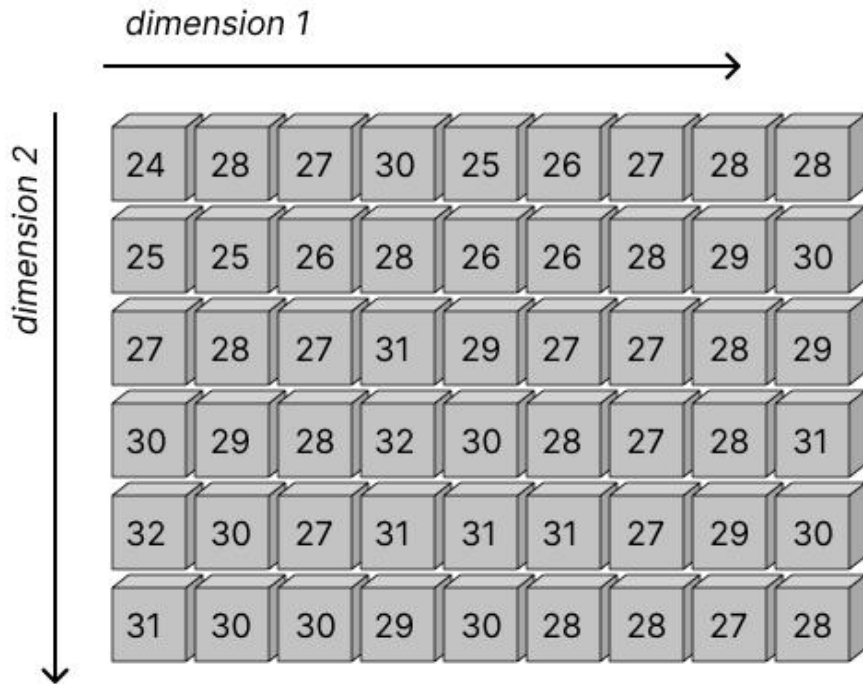
Water temperatures over time at a buoy is an example of one dimensional scientific data



One dimensional



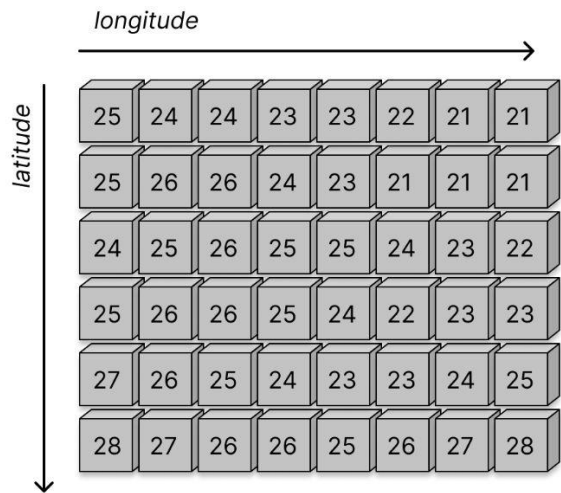
Multi-dimensional data



Two dimensional

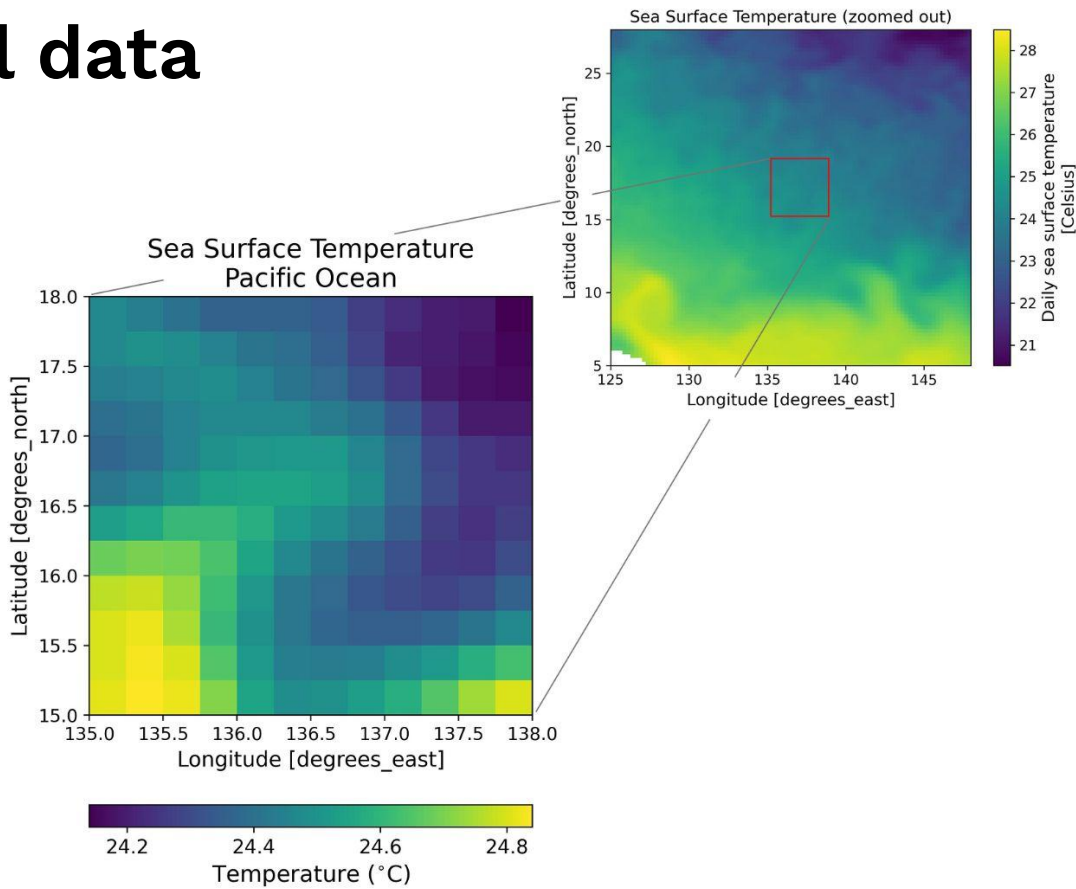
Shape: (9, 6)

Multi-dimensional data

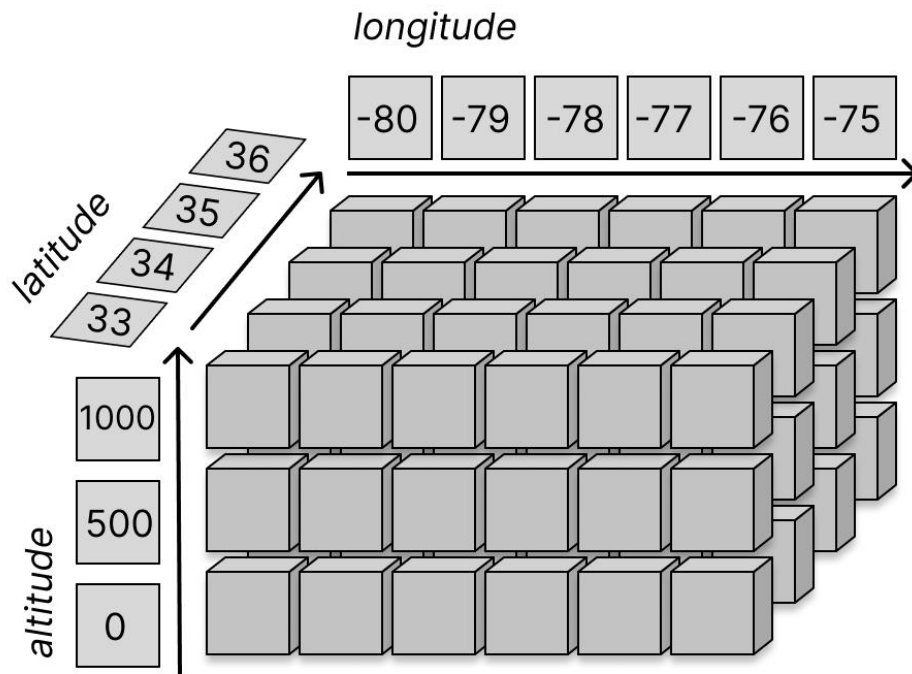


Two dimensional

Shape: (9, 6)



Multi-dimensional data



Three dimensional

Shape: (6, 4, 3)

Xarray DataArray

```
import xarray as xr
```

Type of data structure xarray DataArray Output

xarray.DataArray (latitude: 5, longitude: 6)

Dimensions and their sizes

```
array([[80, 80, 81, 83, 82, 80],  
       [81, 80, 81, 83, 83, 82],  
       [81, 81, 83, 84, 84, 83],  
       [79, 80, 83, 85, 85, 85],  
       [78, 77, 80, 85, 86, 87]])
```

**Array of values
(numpy ndarray)**

▼ Coordinates:

latitude	(latitude)	int64	36 37 38 39 40
longitude	(longitude)	int64	-80 -79 -78 -77 -76 -75

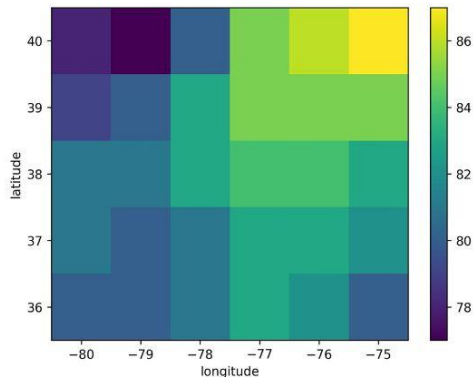
► Indexes: (2)

► Attributes: (0)

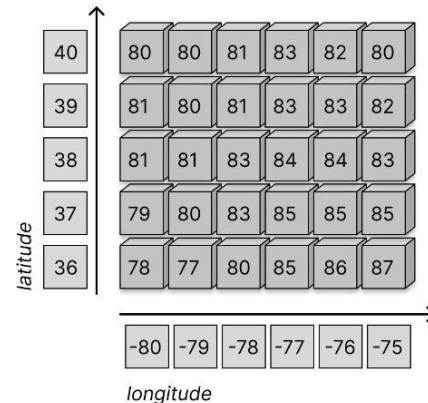
Coordinates (the actual values of the dimensions)

Additional metadata

Data Plotted with a Colorbar



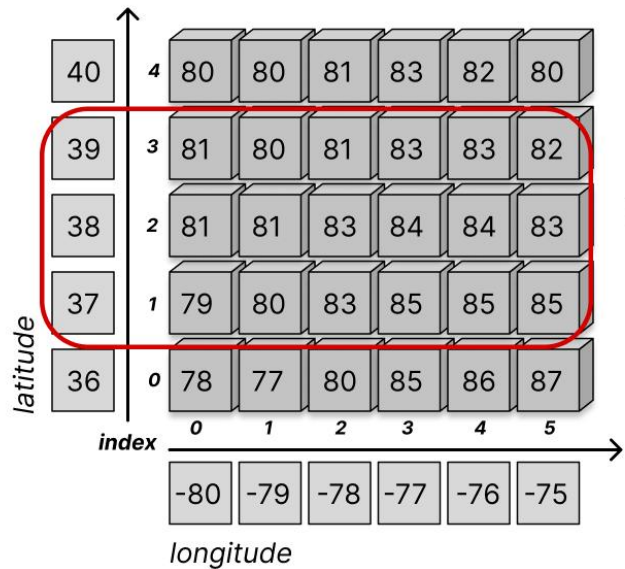
Conceptual Drawing



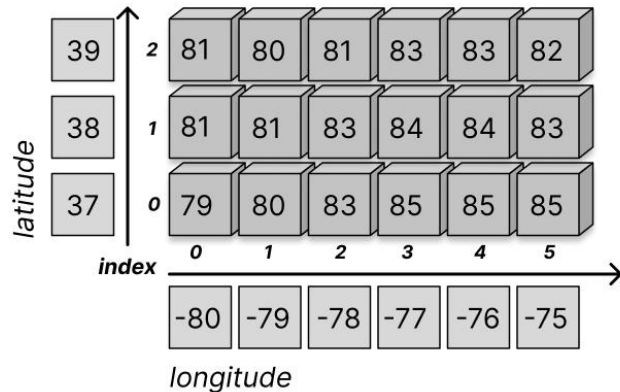
xarray provides Python data structures for working with labelled, multidimensional data



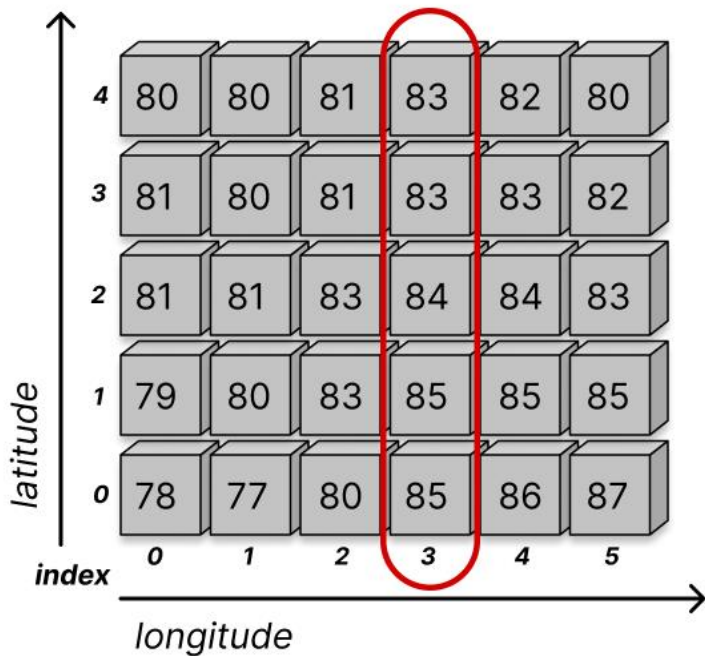
Indexing DataArrays



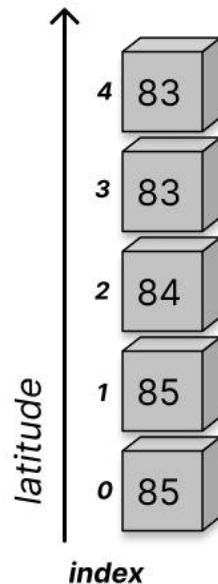
`temps.sel(latitude=slice(37, 39))`



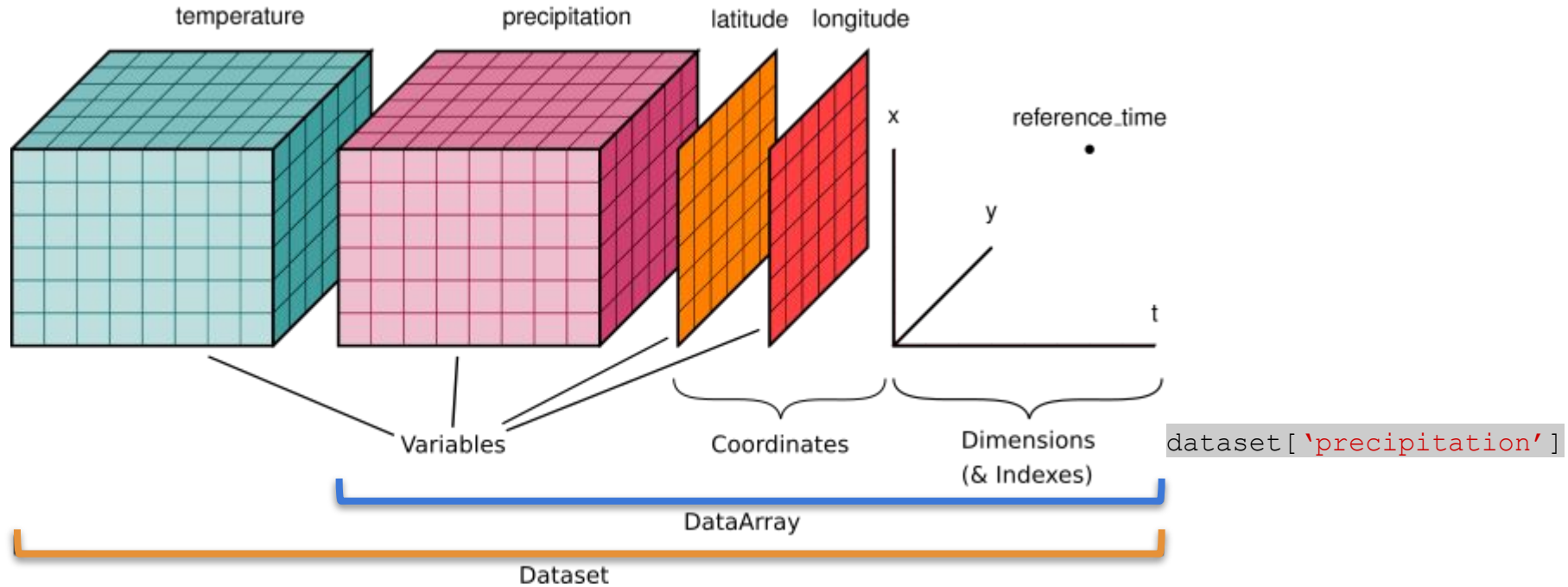
Indexing DataArrays



`temps.isel(longitude=3)`



Xarray DataSet



Xarray DataSet

DataSet

xarray.Dataset

Dimensions: (latitude: 5, longitude: 6)

Coordinates:

latitude	(latitude)	int64	36 37 38 39 40
longitude	(longitude)	int64	-80 -79 -78 -77 -76 -75

Data variables:

temperature	(latitude, longitude)	int64	80 80 81 83 82 ... 77 80 8...
wind_speed	(latitude, longitude)	float64	5.585 12.89 11.65 ... 9.55...

Indexes: (2)

Attributes: (0)



DataArray

```
air_dataset['temperature']
```

[26]:

xarray.DataArray 'temperature'(latitude: 5, longitude: 6)

```
array([[80, 80, 81, 83, 82, 80],
       [81, 80, 81, 83, 83, 82],
       [81, 81, 83, 84, 84, 83],
       [79, 80, 83, 85, 85, 85],
       [78, 77, 80, 85, 86, 87]])
```

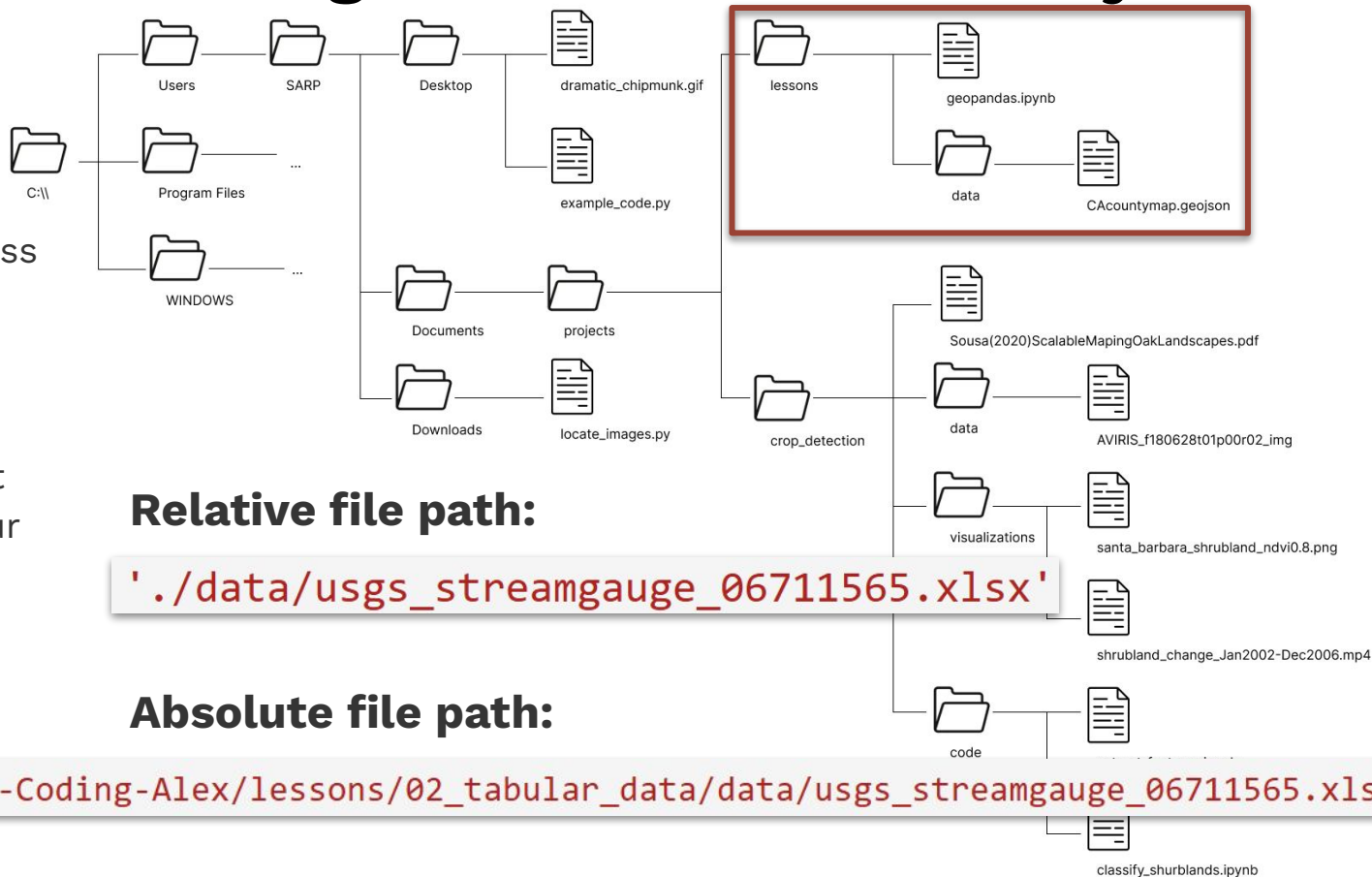
Coordinates:

latitude	(latitude)	int64	36 37 38 39 40
longitude	(longitude)	int64	-80 -79 -78 -77 -76 -75

Indexes: (2)

Attributes: (0)

Computers are organized “hierarchically”



Filepath: An address that specifies the location of any file on your computer

Root: The folder at the very top of your tree that contains everything

Relative file path:

`'./data/usgs_streamgauge_06711565.xlsx'`

Absolute file path:

`'/Projects/2026-Coding-Alex/lessons/02_tabular_data/data/usgs_streamgauge_06711565.xlsx'`

Lesson 4: Gridded Data with xarray

Lesson 4: Gridded Data with xarray

Please complete the **exit ticket**:

<https://forms.gle/UuvpwSEoqDbV5oGo9>

Don't forget to **shut down CryoCloud**:

File → Hub Control Panel → Stop Server